

LESSON 137 (2.0) Ground

Lesson Objective

The purpose of this lesson is to introduce human factors, night operations, human factors relevant to night flying, night VFR cross-country planning, procedures, and regulations applicable to a night VFR cross-country flight. This lesson will also serve as a review on lost procedures and weather information.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Aviation Weather Services Handbook ([FAA-H-8083-25C](#))
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 3 - Airspace.
 - Chapter 6 - Emergency Procedures.
 - Chapter 7: Safety of flight
 - Section 1: meteorology
 - Chapter 8: Medical facts for pilots
- Federal Aviation Regulations.
 - 14 CFR 61 & 91
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 12: Weather Theory
 - Chapter 13: Aviation Weather Services
 - Chapter 16: Navigation
 - Chapter 17: Aeromedical Factors

Instructor Guidance

- Emphasize that many human factor accidents occur at night due to reduced visual cues and increased spatial disorientation risk. Encourage students to rely more on instruments and maintain strong situational awareness
- Let your student know to bring a flashlight (white light for pre-flight & red light for inside the flight deck). Additionally, to verify in-flight lighting is operational e.g. switch lights and panel lights.
- When discussing weather considerations at night, emphasize that haze, fog, and deteriorating visibility can be more difficult to detect.

Lesson Content

Human Factors

- Hypoxia - lack of oxygen
 - Hypoxic Hypoxia - insufficient oxygen available to the body.
 - Caused by insufficient oxygen available to the lungs, usually due to high altitude or inadequate oxygen supply.
- Hypemic Hypoxia - blood is unable to transport oxygen.
 - Inability to absorb oxygen.
 - Caused by CO poisoning, blood loss, anemia.
- Stagnant Hypoxia - blood is not flowing to move oxygen.
 - Caused by reduced blood flow, which prevents oxygen from reaching tissues.
 - High G forces
 - Cold temperatures
- Histotoxic Hypoxia - inability of cells to use oxygen.
 - Caused when cells cannot use oxygen, commonly due to alcohol, drugs, or certain toxins.
- Symptoms of hypoxia (these will differ from person to person)
 - Cyanosis (blue nails and lips).
 - Headache.
 - Impaired judgment.
 - Takes longer to react.
 - Euphoria.
 - Visual impairment.
 - Drowsiness.
 - Lightheadedness or dizziness.
 - Tingling in fingers or toes.
 - Numbness.
- Corrective actions:
 - Use supplemental oxygen (requirements in [91.211](#)).
 - Decrease altitude.
 - Use medication.
 - Don't drink and fly.
 - Stretching – G suits.
- Hyperventilation – excessive rate and depth of respiration that leads to the abnormal loss of carbon dioxide from the blood or too much oxygen. Caused by anxiety, fear, or stressful situations.
 - Symptoms (can be similar to hypoxia):
 - Increased breathing rate.
 - Unconsciousness.
 - Visual impairment.
 - Lightheadedness or dizziness.

- Tingling feelings.
 - Hot/cold sensations.
 - Muscle spasms.
 - Corrective actions:
 - Breathing normally.
 - Breath into paper bag.
 - Talking aloud.
- Middle ear and sinus problems – gasses trapped within the body expand with altitude causing pain in the middle ear and sinuses if gasses are not released. Usually caused due to change in pressure from climbing and descending with blockage.
 - Causes:
 - Illness and blockages.
 - Ear:
 - Pressure builds inside the middle ear – needs to be relieved when at altitude or when back on the ground.
 - Symptoms:
 - Can have loss of hearing.
 - Eardrum could rupture.
 - Infections.
 - Corrective actions:
 - Valsalva maneuver.
 - Yawning.
 - Swallowing.
 - Sinuses (most frequently experienced during descent):
 - Symptoms:
 - Pain in sinuses.
 - Toothache.
 - Bloody mucus discharge.
 - Corrective actions:
 - Slow descent.
 - Don't fly.
 - The Eustachian tube connects the middle ear to the throat and helps equalize pressure.
- Spatial disorientation - lack of orientation in regard to position, altitude, or movement of the airplane (confusion of sensory organs).
 - 3 systems:
 - Vestibular – organs in the inner ear that sense position by the way they are balanced.
 - 3 semicircular canals (based on 3 axes of motion) with fluid and tiny hairs that sense which way fluid is moving in the ear:
 - Horizontal.
 - Superior.

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- Posterior.
- Illusions:
 - The leans – Once in a constant bank, (usually for 30 seconds), the vestibular system will “catch up” with the plane and the pilot will sense that the aircraft is straight and level.
 - Coriolis illusion – after banking for a while and making a sudden head movement, it can give the feeling of a maneuver or turn not occurring. Pilots might make wrong corrections and make things worse.
 - Graveyard spiral – prolonged coordinated turn can feel straight and level, could be due to the leans, (Airspeed increases while the aircraft descends in a tightening spiral.). The pilot will sense the altitude loss and add elevator pressure. But because the aircraft is actually banked, the turn will steepen, and the nose will drop more. Leads to an uncontrolled spiral dive.
 - Graveyard spin – enter a spin which is maintained and felt to decrease, so the opposite rudder is applied and then it feels like a spin in the opposite direction which will be corrected by use of rudder and then a spin in the original direction.
- Somatosensory – nerves in the skin, muscles, and joints, which, along with hearing, sense position based on gravity, feeling and sound.
 - Illusions: Somatogavic illusion – rapid acceleration gives the feeling of being nose-up. Pilots may push the nose down unnecessarily.
 - Inversion illusion – abrupt change from climb to straight and level can give the feeling of tumbling backwards. Pilot may push the aircraft nose lower.
 - Elevator illusion – abrupt vertical acceleration may give false feeling of being in climb causing pilot to push nose lower.
- Visual – eyes, which sense position based on what is seen.
 - Illusions:
 - False horizon – clouds or other obscured horizons can give the illusion of a false horizon. This can lead to the pilot using the wrong horizon reference line.
 - Runway width - A narrower-than-usual runway can create an illusion that the aircraft is at a higher altitude than it actually is. This can lead to the pilot flying a lower approach or rounding out too late. A wider runway can have the opposite effect with the risk of the pilot leveling out the aircraft high and landing hard or overshooting the runway.
 - Runway slope - an upsloping runway can create an illusion that the airplane is higher than it actually is. This can lead to the pilot flying a lower approach. A down sloping runway can create an illusion that the airplane is lower than it actually is. This can lead to the pilot flying a higher than normal approach.
 - Featureless terrain/Black Hole approach - An absence of ground

features (like over the water, dark areas, desert, etc.) can create an illusion that the airplane is higher than it actually is. This can lead to the pilot flying lower or a lower approach.

- What is motion sickness?
 - Caused by the brain receiving conflicting messages about the state of the body.
 - Symptoms:
 - Loss of appetite.
 - Dry mouth.
 - Sweating.
 - Nausea.
 - Headaches.
 - Corrective actions:
 - Open air vents.
 - Focus on objects outside of the airplane and in the distance.
 - Avoid unnecessary head movements.
 - Avoid greasy and/or heavy meals before flying.
- What is carbon monoxide poisoning?
 - Colorless, tasteless, odorless gas that prevents hemoglobin from carrying oxygen to the cells.
 - Causes
 - Inhalation of combustion fumes.
 - Cars, planes, charcoal grills.
 - Symptoms
 - Headache.
 - Drowsiness.
 - Dizziness.
 - Corrective actions
 - Shut off heater.
 - Open air vents.
 - Seek medical attention.
- Fatigue and Stress
 - Fatigue
 - Acute – felt after long periods of physical strain.
 - Short term.
 - Coordination and alertness may be reduced.
 - Prevented by adequate sleep, exercise, nutrition.
 - Chronic fatigue - occurs by not recovering from acute fatigue.
 - Long term.
 - Performance can decrease.
 - Prolonged periods of rest are needed.

- Stress:
 - Body's response to demands placed on it.
 - Physical and psychological.
 - Small levels are ok.
 - Can be a burst of energy/motivation.
 - Avoid by exercising, well nourished, and well rested.
- Dehydration and heatstroke – lack of fluids for the body to carry for normal functions.
 - Dehydration – occurs by inadequate intake, or loss of, fluids. Drink plenty of fluids.
 - Avoid caffeine and alcohol intake.
 - Symptoms:
 - Headaches.
 - Nausea.
 - Internal organs can shut down.
 - Can lead to heatstroke .
 - Heatstroke – body is unable to keep itself cool.
 - Similar symptoms to dehydration but can lead to complete collapse.
- Effects of alcohol and drugs – FAR [91.17](#):
 - Never combine drugs, alcohol, and flying.
 - 1 Even small amounts of alcohol can impair judgment, coordination, and reaction time.
 - Maximum .04 BAC.
 - Can increase the effects of hypoxia and dehydration.
 - FAA prohibits cockpit operations if you consume alcohol within 8 hours (12 hours for FIT).
 - If medication must be taken, ensure it is approved by the FAA (ask AME).
- Effects of nitrogen excess during scuba dives
 - Nitrogen is absorbed into the bloodstream during dives due to increased pressure and can expand and cause severe pain.
 - After uncontrolled ascent:
 - Wait at least 12 hours to fly up to 8,000 ft. after diving.
 - Wait at least 24 hours before flights above 8,000 ft.
 - After controlled ascent:
 - Wait 24 hours before flying at any altitude.
- Types of decompression sickness – nitrogen bubbles.
 - Bends – joints, elbows, shoulders, lips.
 - Chokes – in the lungs.
 - Skin – itching on the skin, swelling and skin compressions.
 - Neurologic – Brain, spinal cord, and peripheral nerves.
 - Recovery techniques:
 - Land as soon as possible.
 - Declare emergency if needed.

- Barometric chamber if needed.
- Human Factors Relevant to Night Flying

Visual Scanning and Collision Avoidance

- Quiz the student on how to scan and avoid traffic.
- Quiz the student on right of way rules.
- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Regulations Relevant to Night Flying

- Several regulations in 14 CFR Parts 61 and 91 apply specifically to night flying. Pilots must understand these requirements when planning and conducting night operations.
- 14 CFR § 61.57(b) – Night Passenger Carrying Currency
To carry passengers at night, a pilot must have completed three takeoffs and three landings to a full stop during the preceding 90 days. These takeoffs and landings must be performed during the period beginning 1 hour after sunset and ending 1 hour before sunrise.
- 14 CFR Appendix-B-to-Part-141 4.(b)(1)(ii)(A) & (B)– Night Training Requirements for Private Pilot
Private pilot applicants must receive at least 3 hours of night flight training, which includes:
 - One night cross-country flight of over 100 nautical miles total distance.
 - 10 takeoffs and 10 landings to a full stop at an airport with an operating control tower.
- 14 CFR § 91.205(c) – Required Equipment for Night VFR Flight
In addition to the equipment required for day VFR flight, aircraft operated at night must have the following equipment (FLAPS):
 - **F**uses or circuit breakers
 - **L**anding light (required if the aircraft is operated for hire)
 - **A**nti-collision lights
 - **P**osition lights
 - **S**ource of electrical power (alternator or generator)
- 14 CFR § 91.151 – Fuel Requirements for VFR Flight
For night VFR operations, no person may begin a flight unless the aircraft has sufficient fuel to fly to the first point of intended landing and then fly for at least 45 minutes at normal cruising speed.
- 14 CFR § 91.155 – Basic VFR Weather Minimums
Pilots must comply with the applicable visibility and cloud clearance requirements when operating under VFR conditions at night.

VFR Cross-Country Flight Planning (Night)

- Night cross-country flight planning follows the same general procedures as daytime planning but requires additional consideration for visibility, terrain, lighting, and emergency options. Pilots must carefully evaluate conditions and plan the route to ensure safe navigation in reduced lighting environments.
- Before departure, the pilot should review:
 - Weather Information
 - Obtain a complete weather briefing, including METARs, TAFs, winds aloft, NOTAMs, and radar imagery.
 - Pay special attention to visibility, cloud ceilings, and potential weather deterioration, as visual cues are more limited at night.
 - Be cautious of haze, fog, or low clouds, which are more difficult to detect in darkness.
 - Route Selection
 - Plan routes that follow well-lit landmarks or populated areas when possible.
 - Avoid flying over large dark areas such as forests, water, or sparsely populated terrain, where visual references may be limited.
 - Consider terrain and obstacle clearance carefully since terrain can be more difficult to identify at night.
 - Fuel Planning
 - Follow 14 CFR § 91.151, which requires sufficient fuel to reach the first point of intended landing plus 45 minutes reserve for night VFR flight.
 - Airport and Lighting Considerations
 - Verify that departure, destination, and alternate airports have operational runway lighting.
 - Review lighting systems such as pilot-controlled lighting (PCL) and ensure familiarity with the required radio frequencies to activate the lights.
 - Review airport diagrams and lighting information in the Chart Supplement.
 - Emergency Planning
 - Identify suitable alternate airports along the route of flight.
 - Plan routes that allow access to lighted areas or airports in the event of an emergency.
 - Be aware that finding suitable landing areas is significantly more difficult at night.
 - Equipment Check
 - Ensure all required night VFR equipment (FLAPS) is operational in accordance with 14 CFR § 91.205(c).
 - Verify that cockpit lighting, flashlights, and navigation equipment are functioning properly.

Lost Procedures

- 7C's:
 - Calm
 - Climb

- Identify prominent landmarks.
- Circle.
 - Avoid aggravating the situation.
- Crosscheck
 - GPS.
 - VOR: use 2 VOR's, center the CDI, draw on the sectional chart to determine position.
- Communicate
 - Request assistance from ATC.
- Comply
 - Follow ATC instructions.
- Conserve
 - reduce power to 55% and lean the mixture for best economy (refer to PIM).

Weather Information

- Review from Lesson 131

Emergency Equipment and Survival Gear

- Review from Lesson 131

Systems and Equipment Malfunctions

- Review from Lesson 131

Completion Standards

This lesson is complete when the applicant:

1. Has been introduced to human factors, night operations, and night cross-country planning and procedures.
2. Can plan a night cross-country flight with instructor guidance.
3. Shows proficiency in lost procedures and weather information.